MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCPHY201 -
ELECTROMAGNETIC THEORY-SCHEME OF VALUATION
(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Part A
Short Answer
Answer any 10 questions
Each question carries 3 marks

1. Calculate the work done in moving a charge of 1 C around the circumference of a circle of radius R, starting and ending at the same point on the circle. [U]/[CO1]
Ans: Electrostatic field conservative – 2
Work done in closed loop = 0 – 1
2. State Gauss's law in differential form and its relation to divergence. [U]/[CO1]
Ans: $\nabla \cdot E = \rho/\varepsilon_0$ – 2
Relation to divergence – 1
3. Can Gauss law be applied to non-uniform charge distributions? Justify your answer briefly. [U]/ [CO1, CO2]
Ans: Yes – 1
General validity and Explanation – 2
4. What is electrostatic shielding? [U]/ [CO2]
Ans: 3 marks
(Definition alone – 1, Example alone – 1)
5. What is the current in a line of charge? [U]/[CO3] Ans:
Ans: Definition and expression – 3
(Expression alone - 1, Definition alone -1)
6. State the Coulomb gauge condition for vector potential in magnetostatics. [U]/[CO3]
Ans: Explanation - 3 marks (mention no monopole only ‑ 1, $\nabla \cdot B = 0$ only – 1)
7. Why are magnetic field lines always closed loops? [U]/ [CO3, CO4]
Ans: Explanation - 3 marks (mention no monopole only ‑ 1, $\nabla \cdot B = 0$ only – 1)
8. What is the permittivity of free space? [U]/[CO5]
Ans: Description or value with unit - 3 marks (Value alone– 1, Unit alone – 1)
9. Define the term magnetization. Suggest a method to enhance the magnetization in materials. [U]/[CO5]
Ans: Definition or Relation and Method – 3 marks
(M relation alone – 1, Method alone – 1)
10. At what condition a neutral atom have an induced electric dipole moment? [U]/[CO5]
Ans: Explanation - 3 marks (Polarization alone – 1, External field alone– 1)
11. Write Faraday's law in differential form and explain what it tells us about the curl of the electric field. [U]/[CO6]
Ans: $\nabla \times E = -\partial B/\partial t$ – 2
Curl meaning – 1
12. What are free charge and bound charge in matter? [U]/[CO7]
Ans: Definitions – 1.5 each

[3 × 10 = 30]

Part B

Short Essay

Answer any 8 questions

Each question carries 5 marks

13. Derive Poisson's equation and explain its physical significance. [U]/ [CO1, CO2]

Ans: Gauss law – 1
E=-∇V –1
Derivation –2
Significance –1

14. A charge of 6μC is placed at a point in an electric field of strength 3 × 10^4 N/C. Find the force experienced by the charge. [A]/ [CO2]

Ans: F = qE -2
Substitution –2
F = 0.18 N final answer with unit -1

15. Explain why the excess charge resides on the surface of a conductor. [U]/ [CO2]

Ans: Free charge –1
Repulsion –1
Surface reason –2
Conclusion –1

16. Differentiate between magnetic induction and magnetic flux. [U]/ [CO3]

Ans: Definitions –2
Units –1
Differences –2

17. At what distance from a 20 A current-carrying wire will the magnetic field be 4 × 10^-5 T? [A]/ [CO4]

Ans: $r = (\mu_0 I) / (2\pi B)$-2
Substitution -2
Final Answer with unit -1
r = 0.1 m = 10 cm Formula –2

18. Explain magnetic susceptibility and discuss its role in characterizing magnetic materials. [U]/ [CO5]

Ans: Definition –2
Relation –1
Role –2

19. The potential inside a uniformly polarized sphere is $V(r, \theta) = (P\ r \cos\theta)/(3\varepsilon_0)$. What is the electric field inside the sphere? [E]/ [CO5]

Ans: E=-∇V –2
Differentiation –2
Final –1

20. State and explain Gauss's law in dielectric media with mathematical form. [U]/ [CO5]

Ans: Statement –2
D field –2
Equation –1

21. Define electromotive force. Derive an expression for the terminal voltage of a cell in terms of emf, current, and internal resistance. [U]/ [CO6]

Ans: Definition –1
V=E−Ir –3
Explanation –1

22. Explain how a changing magnetic field produces an electric field. [A]/ [CO7]

Ans: Faraday law –2

Curl relation –2
Explanation –1